

SLIIT
SE3082 – Parallel Computing



Lab 7
Exercise 06

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Exercise 06: Scatter + Scan

Implementation Code (sum_scan.c)

```
#include <mpi.h>
#include <stdio.h>
#include <stdlib.h>
#define N 1000000

int main(int argc, char **argv) {
    MPI_Init(&argc, &argv);

    int rank, size;

    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    int chunk_size = N / size;

    int *array = NULL;

    if (rank == 0) {
        array = (int *)malloc(N * sizeof(int));
        for (int i = 0; i < N; i++) array[i] = i + 1;
    }

    int *local_chunk = (int *)malloc(chunk_size * sizeof(int));

    MPI_Scatter(array, chunk_size, MPI_INT, local_chunk, chunk_size, MPI_INT, 0,
        MPI_COMM_WORLD);

    long long local_sum = 0;
    for (int i = 0; i < chunk_size; i++) local_sum += local_chunk[i];

    long long prefix_sum = 0;

    MPI_Scan(&local_sum, &prefix_sum, 1, MPI_LONG_LONG, MPI_SUM, MPI_COMM_WORLD);

    long long sum_before_me = prefix_sum - local_sum;

    printf("Rank %d: local_sum = %lld, prefix_sum = %lld, sum_before_me = %lld\n", rank, local_sum,
        prefix_sum, sum_before_me);

    if (rank == 0) free(array);

    free(local_chunk);

    MPI_Finalize();
}
```

```
return 0;  
}
```

Test Output (4 processes)

Rank 0: local_sum = 31250125000, prefix_sum = 31250125000, sum_before_me = 0

Rank 1: local_sum = 93750125000, prefix_sum = 125000250000, sum_before_me = 31250125000

Rank 2: local_sum = 156250125000, prefix_sum = 281250375000, sum_before_me = 125000250000

Rank 3: local_sum = 218750125000, prefix_sum = 500000500000, sum_before_me = 281250375000